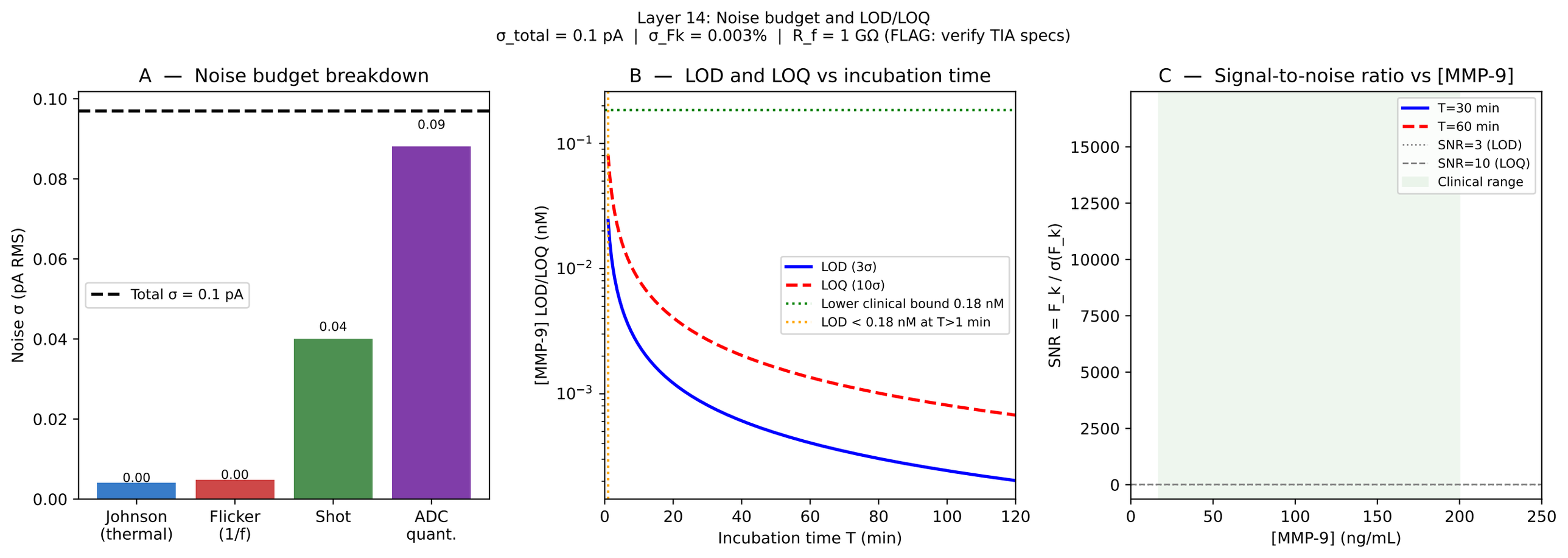
Layer 14: Noise budget and LOD/LOQ
σ_total = 0.1 pA | σ_Fk = 0.003% | R_f = 1 GΩ (FLAG: verify TIA specs)
A — Noise budget breakdown
Noise σ (pA RMS)
Total σ = 0.1 pA
0.00
0.00
0.04
0.09
Johnson (thermal)
Flicker (1/f)
Shot
ADC quant.
B — LOD and LOQ vs incubation time
[MMP-9] LOD/LOQ (nM)
LOD (3σ)
LOQ (10σ)
Lower clinical bound 0.18 nM
LOD < 0.18 nM at T>1 min
Incubation time T (min)
C — Signal-to-noise ratio vs [MMP-9]
SNR = F_k / σ(F_k)
T=30 min
T=60 min
SNR=3 (LOD)
SNR=10 (LOQ)
Clinical range
[MMP-9] (ng/mL)